	Sultanate of Oman Ministry of Health The Royal Hospital Department of Child Health

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Title: Malaria Management in Paediatric

Statement:

This policy is made to guide Pediatrician in the management of malaria.

Introduction:

- Malaria is an imported parasitic infection transmitted to humans by the bite of female Anopheles mosquitoes and is caused by five species of *Plasmodium - falciparum, vivax, ovale, malariae or knowlesi*
- *P. falciparum* is the most common cause of severe malaria which is a medical emergency but *P. knowlesi,* and more rarely *P. vivax* can also cause severe illness.
- *Incubation period of malaria between 7days and 3 months*

General comments:

◊ All malaria cases should be notified within 24 hours

◊ Availability of antimalarial medicines is limited only to government health facilities

◊ Management of malaria cases is free of charge

Transmission of malaria:

It transmitted by the bite of an infective female Anopheles mosquito. Occasionally, transmission occurs by blood transfusion, organ transplantation, needle sharing, or congenitally from mother to fetus.

Investigations required for suspected malaria patient:

- CBC, LFT, UE1, serum glucose, G6PD and HPLC.
- Thick &thin smear, Parasitic index: Estimate the percentage from the smear
- If severe malaria adds to above investigation: LFT, LDH, Coagulation, (Blood gas (if resp distress).

Management of uncomplicated malaria:

1. Other species than falciparum malaria species
2. > 5 years of age.
3. Clinically well and has been reviewed by a senior and experienced clinician.
4. No clinically or laboratory features of severe malaria.
5. Parasitemia < 2%.
6. Tolerate first dose of oral medication(observed post administration for 4-6 hrs)
7. Has resided in an endemic area for most of the previous year.
8. Family has sufficient understanding to ensure compliance and follow up
9. Family has sufficient understanding to know to return to the hospital immediately if the child becomes unwell.
10. Child has no other co morbidity necessitating admission.

Treatment of uncomplicated malaria

Treatment of uncomplicated malaria **P falciparum or** **unidentified species**	**Antimalarial treatment:** **Artemether + lumefantrine:** 1 tablet=20 mg artemether/120 mg lumefantrine Therapeutic dose: 6 doses within 3 days Orally, at 0, 8, 24, 36, 48, and 60 hours 5–14 kg: 1 tablet 15–24 kg: 2 tablets 25–34 kg: 3 tablets >34 kg: 4 tablets (adult dose) Should be taken after a meal or drink containing at least 1.2 g fat – particularly on the 2nd and 3rd day of treatment. **Primaquine (anti-gametocytes)**: a single dose of 0.25 mg/kg bw primaquine should be given (except pregnant women, infants aged < 6 months and women breastfeeding infants aged < 6 months) to reduce transmission. G6PD testing is not required. **Alternative treatments:** - Quinine sulfate plus one of the following: - Clindamycin or if the child >8 years Doxycycline, Tetracycline - Quinine sulfate: 8.3 mg base/kg (=10 mg salt/kg) PO TID x 3 or 7 days. - Clindamycin: 20 mg base/kg/day po divided TID x 7 days. - Doxycycline (>8 years old): 2.2 mg/kg po 12 hourly x 7days. - Tetracycline ((>8 years old):25 mg/kg/day po divided QID x 7days.

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<td>Management of uncomplicated malaria caused by P vivax, ovale and malariae</td>
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have high prevalence of chloroquine-resistant

· Rare cases of chloroquine-resistant P. vivax have also been documented in Burma (Myanmar), India, and Central and South America.
· Persons acquiring P. vivax infections from regions other than Papua New Guinea or Indonesia should initially be treated with chloroquine.

In areas with chloroquine-sensitive P.vivax / P.ovale, P.malariae:

Chloroquine is given at an initial dose of 10 mg base/kg body weight, followed by 10 mg/kg body weight on the second day and 5 mg/kg body weight on the third day

In areas with chloroquine-resistant P. vivax

Artemether + lumefantrine:

1 tablet=20 mg artemether/120 mg lumefantrine Therapeutic dose: 6 doses within 3 days
Orally, at 0, 8, 24, 36, 48, and 60 hours
5–14 kg: 1 tablet

15–24 kg: 2 tablets
25–34 kg: 3 tablets
>34 kg: 4 tablets (adult dose)
Should be taken after a meal or drink containing at least 1.2 g fat – particularly on the 2nd and 3rdday of treatment.

<u>Liver hypnozoite eradication for P ovale and vivax malaria:</u>

-Only give after check G6PD

If G6PD normal :

To prevent relapse, an additional treatment option of using primaquine 0.5 mg/kg/day for seven days is recommended to treat P. vivax or P. ovale malaria in children and adults (except pregnant women, infants aged < 6 months, women breastfeeding infants aged < 6 months, women breastfeeding older infants unless they are known not to be G6PD deficient, and people with G6PD deficiency).
In people with G6PD deficiency, primaquine base at 0.75 mg/kg body weight once a week for 8 weeks can be given to prevent relapse, with close medical supervision for potential primaquine-induced haemolysis.

<u>Primaquine is contraindicated:</u>

1-Severe G6PD deficiency (not active G6PD)

2-Less than 6 months of age

3- Primaquine is contraindicated in pregnant women, infants (< 6 months) and conditions predisposing to granulocytopenia, including active rheumatoid arthritis and lupus erythematosus
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- Monitoring therapeutic efficacy for plasmodium vivax and P.Falciparium: repeat malaria smear on day 28.

Severe malaria which includes any of the following clinical features or lab findings:

Clinical features:

- Impaired consciousness (may indicate hypoglycemia and/or cerebral malaria)
- Prostration, i.e. generalized weakness so that the patient is unable walk or sit up without assistance
- Failure to feed
- Multiple convulsions – more than two episodes in 24 h
- Deep breathing, respiratory distress (acidotic breathing), or pulmonary edema or Acute respiratory distress syndrome
- Hyperpyrexia (fever >40^0C).
- Circulatory collapse or shock, (BP < 50 mm Hg) in children
- Jaundice with end organ dysfunction (shock or failure)
- Anemia (Hb < 5g/dl).
- Spontaneous bleeding (may indicate disseminated intravascular coagulation - DIC).

Laboratory findings:

- Hypoglycemia (blood glucose < 2.2 mmol/l or < 40 mg/dl).
- Metabolic acidosis (plasma bicarbonate < 15 mmol/l).
- Severe normocytic anemia (Hb < 5 g/dl, packed cell volume < 15%).
- Hemoglobinuria (urine test positive for blood without red blood cells seen on microscopy).
- hyperparasite (> 2%/100 000/µl in low intensity transmission areas or > 5% or 250 000/µl in areas of high stable malaria transmission intensity: Parasite density should be monitored every 12 hours)
- hyperlactatemia (lactate > 5 mmol/l)
- Renal impairment (serum creatinine > 265 µmol/l)

Management of severe malaria

Severe malaria: **any species (P falciparum,P vivax,P ovale,P malariae,P knowlesi**	• IV Artesunate 2.4 mg/kg IV at 0, 12, and 24 hours, then once daily (IV treatment should continue at least 24 hours) until oral therapy is tolerated • Children weighing < 20 kg should receive a higher dose of artesunate (3 mg/kg body weight per dose) than larger children and adults (2.4 mg/kg body weight per dose) to ensure equivalent exposure to the drug. • Alternative treatment (if artesunate not available): • IV Quinine dihydrochloride plus one of the following: Doxycycline, Tetracycline, or Clindamycin (doses as mentioned above) - Quinine dihydrochloride Loading dose: 20 mg/kg IV over 4 hours. (usually diluted in 5-10 ml/kg of 5% dextrose saline).

	- Re-check Blood Glucose (BG) every 2-4 hours during the course of IV quinine. - If BG (<2.2 mmol/l OR 40 mg/dl) infuse dextrose over a period of 3-5 minutes. • Loading dose of quinine is not required if patient: -Has received 3 or more doses of quinine or quinidine in the previous 48 hours - Mefloquine prophylaxis in the previous 24 hours mefloquine treatment dose within the previous 3 days. - Maintenance dose: starting 8 hours after initiation of loading dose, IV 10 mg/kg /dose 8 hourly (to be diluted in 5-10 ml/kg 5% dextrose saline, infuse over 4 hours) • IV quinine to be given a minimum of 24 hours, irrespective of the patient's ability to tolerate oral medication earlier or until the patient is about to tolerate oral medication. • **<u>When patient is able to tolerate oral therapy:</u>** give (doses see uncomplicated malaria): • Artemether lumefantrine —full course of 6 doses. OR • Quinine+(clindamycin OR doxycycline)—full 7days course.

- **<u>Consider exchange transfusion</u>** if the parasite density is

> 10% OR if the patient has altered mental status, non-volume overload pulmonary edema, or renal complications.

- Exchange transfusion should be continued until the parasite density is <1% (usually requires 8-10 units)

Exchange Transfusion:

-One and half volume of RBC exchange = 1/12 (hematocrit X Total body volume)

Neonate = wt(kg) X 100ml/kg.

1month-10years =wt(kg)X 80ml/kg

Over 10 years =wt(kg)X 70ml/kg.

<u>Continuation of severe malaria treatment</u>

-IV quinidine administration should not be delayed for an exchange transfusion and can be given concurrently throughout the exchange transfusion.

-less than 5% with severe malaria has DIC → Treat with vitamin K.

- Secondary pneumonia/or sepsis treat empirically with 3rd gen cephalosporine unless admitted with clear evidence of aspiration treat with clindamycin.

- Parasitaemia may increase over the first 24-36 hours (especially in severe malaria) and does not indicate treatment failure or resistance

- Gametocytes (sexual stages) may persist or appear during or after treatment – these do not indicate treatment failure

Treatment Failure:

- If fever and parasitemia failed to resolve or they recur within two weeks of treatment then this is considered a failure of treatment.

- Treatment failures may result from drug resistance, poor adherence or inadequate drug exposure.

- It is important to determine from the patient's history whether he or she vomited the previous treatment or did not complete a full course.

- Treatment failures within 14 days of initial treatment should be treated with a second-line antimalarial.

- An alternative Artemisinin-based combination therapy (*ACT*) known to be effective in the region

- Artesunate plus tetracycline or doxycycline or clindamycin (given for a total of 7 days),
- Quinine plus tetracycline or doxycycline or clindamycin (given for a total of 7 days).

Recrudescence: the situation in which parasitaemia falls below detectable levels and then later increases to a patent parasitaemia. Most recrudescence episodes occur more than 2weeks (between 2-6 weeks) after treatment. This can happen because of :

1. Incorrect dosing
2. Incomplete adherence compliance)
3. Poor drug quality
4. Interactions with other drugs
5. Compromised drug absorption
6. Vomiting of the medicine
7. Unusual pharmacokinetic
8. Misdiagnosis

Malaria Prophylaxis:

Regimens of chemoprophylaxis are used:

1-Chloroquine (weekly) +_Proguanil

2-Mefloquine (weekly).

3-Doxycycline (Daily).**CC use h Africa:**

1-Chloroquine (200mg tablets containing 150 mg base) dose 5mg/kg weekly. To be taken 1 wk before travel into an endemic area and should be continued for 4 wks after leavening the endemic area.

Weight (kg)	Age(years)	Number of tablet
5-6	<4 months	1/4
7-10	4-11 months	1/2
11-14	1-2	1/2

19-24	5-7	1
25-35	8-10	1
36-50	11-13	2
50+	14+	2

Chloroquine contraindicated in patients with epilepsy, known chloroquine hypersensitivity reaction, suffering from psoriasis. The clinician has to check the countries that are resistant to chloroquine to avoid prescribing it to the traveler to those countries.

2-Proguanil (100mg tablets containing 87mg base). To be taken 1 wk before travel into an endemic area and should be continued for 4 wks after leavening the endemic area.

Weight(kg)	**Age(years)**	**Number of tablets**
5-8	< 8 month	1/4
9-16	8mon-3yrs	1/2
17-24	4-7	3/4
25-35	8-10	1
36-50	11-13	**1** 1/2
50+	14+	2

Proguanil contraindicated inpatient with liver or kidney dysfunction.

3-Mefloquine (tablets of 250mg base).To be taken 2-3 wks before travel and should be continued for 4 wks after leavening the endemic area. It not recommended for less than 5 kg.

Weight(kg)	Age(yrs)	Number of tablets
<5	<3 months	NR*
5-12	3-23 months	1/4
13-24	2-7	1/2

36-50+	11-14+	1

Mefloquine contraindicated in patients with neuropsychiatric disease, who received mefloquine treatment within 4 wks.

Mefloquine interaction: possible increase risk of bradycardia with concomitant use of betablockers, Calcium-channel blocker and digoxin.

4-Doxycyline (100mg capsule/tablets)dose:1.5mg/kg

It is contraindicated in children less than 8 years.

References

Title of book/ journal/ articles/ Website	Author	Year of publication	Page
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